Design of Bike Frame

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I. Introduction

The objectives of this project are to optimize the diameter of bike frame segments using FSD (fully stressed design) and buckling analysis in Abaqus modeling software to minimize the weight of the 8 tube segments shown in Figure 1 while supporting given loads and preventing buckling.

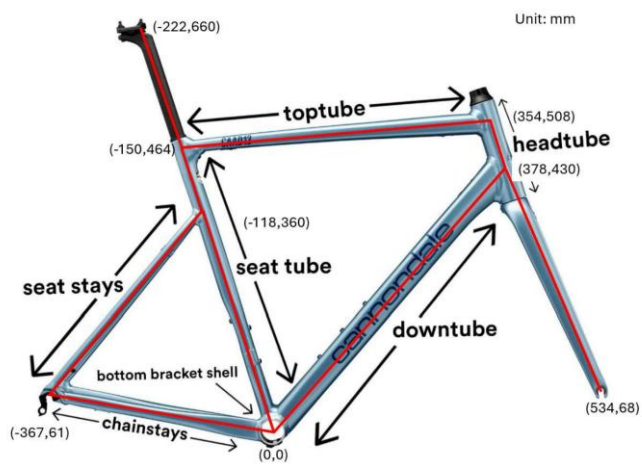

Figure 1 Given geometry of bike frame

For the simplicity of the analysis all the bike tubes will be considered pipe geometries and the loading will be considered in 2 dimensions. The loading considered for the project will be a downward load of 3000N at the seat position, a downward load of 1000N at the pedal crank position and a horizontal load of 1500N at the front wheel location pointing back toward the bike frame simulating a small crash into an object. The dynamic loading will be simulated using a static analysis so a dynamic load factor G of 3 and a safety factor S of 1.5 will be applied to the loading for this project. The boundary conditions for this project will be a pin support at the rear wheel (U1 = U2 = 0) and a horizontally sliding boundary condition (U2 = 0) for the front wheel. Table I describes the units used in this project.

Table I
Table of Units

Quantity	Value	Notes
Length	mm	Length unit in Abaqus
Force	N	Force unit in Abaqus
Stress	MPa	N/mm²
Mass	Kg	Mass unit in Abaqus

The bike frame in this project will be made using Aluminum with the properties listed in Table II.

Table II
Properties of Aluminum

Property	Value	Unit
Density	2580	kg/m³
Young's modulus, E	70,000	MPa
Poisson's ratio, ν	0.33	-
Tube thickness, t	2	mm
Yield Stress	240	MPa

For this project, the diameter of the bike tubes will all start at a set diameter and will be iterated 10 times using the fully stressed design until the diameters of the tubes are optimized for weight while having stress under the yield stress of 240 with the minimum diameter of the tubes being 10 mm. The fully stressed design method or FSD uses the current section modulus of the bike tubes using the following equation:

$$S = \frac{\pi}{32}\frac{{D^4}_0 - {D^4}_i}{D_0} \tag{1}$$

Where D_0 is the outer diameter of the tube and D_i is the inner diameter of the tube. The new section modulus is then calculated from the old section modulus and the max stress in the tube element calculated by Abaqus using the following equation:

$$S_{new} = S_{old}\frac{\sigma_{max}}{\sigma_{allow}} \tag{2}$$

The new section modulus is then used to calculate the new tube diameter by solving equation (1) for D using a nonlinear equation solver. This process is continued for multiple iterations.

Once the tube diameters are optimized for the allowable stress using FSD, a buckling analysis is done using Abaqus to find the eigenvalue or critical load percentage needed for buckling to occur to see if further optimization of the tube

diameters is needed to prevent buckling of the frame below the loading being tested would be reached (eigenvalue < 1).

II. Procedure

For this project, the assumptions made were that all the frame tubes were modeled as straight hollow pipes of constant thickness of 2 mm, the aluminum material was linear elastic and isotropic, local joints, weld effects, and nonlinear contact were neglected, and that the reported mass represents the tubular frame only.

To perform a finite element analysis on the bike frame in Abaqus the bike frame was first modeled in Abaqus using the coordinates of the geometry given in Figure 1. The geometry was then meshed and seeded with 5 elements for every individual tube except for the head tube that was given 3 elements and the seat tube that was given 5 elements to the top portion and bottom portion individually as shown in figure 2.

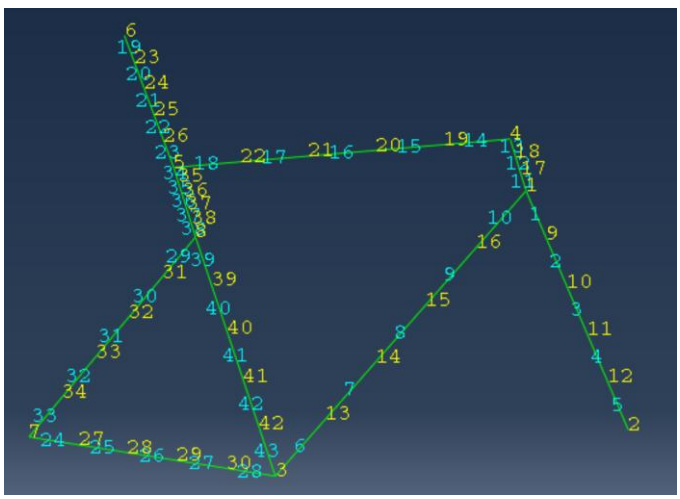

Figure 2 Bike frame modeled in Abaqus with elements labeled in blue and nodes in yellow

Each tube of the frame was given a pipe profile with diameter of 60 mm to start. The boundary conditions to the front and rear wheel were added to the geometry and the load cases multiplied by the dynamic load factor G and the safety factor S were added to the model. Once the model was finalized a job was created and submitted giving the FEA results for the beam stress (S11, maximum of bottom and top surfaces).

Using the first max stress numbers the FSD process for iterating new diameters could be done. The section modulus S for the initial diameter of all the tubes was calculated using equation (1) and then using the max S11 stress reported from Abaqus for each tube, the new section modulus for each tube were calculated using equation (2). The new outer diameter was then calculated using a nonlinear equation solver in MATLAB from equation (1). The new diameters were then put into Abaqus for each tube profile and a new job was created and ran. Using the new stresses from the new job, the process of iterating the diameters of the tubes using the FSD continued optimizing each tube diameter to be their minimum size while getting closer to the yield stress of 240. After 10 iterations of the diameter of the tube all of the S11 stresses should be under 240.

After minimizing the tube sizes using FSD until all of the tube stresses are as high as they can be without being over 240 a buckling analysis is conducted on the bike frame. A buckle step is added to the model reporting 5 eigenvalues or modes of buckling. A new job is created and submitted to get the eigenvalue for the first mode of buckling. If the eigenvalue for the first buckling mode is under 1 then the diameter of the buckled tube is heuristically increased and the buckling analysis is resubmitted. This process is continued until the first buckling mode reports an eigenvalue greater than 1.

III. Results

The final FE (Finite element) geometry for the bike frame in Abaqus with seeded elements, applied loads, and boundary conditions are shown in figure 3.

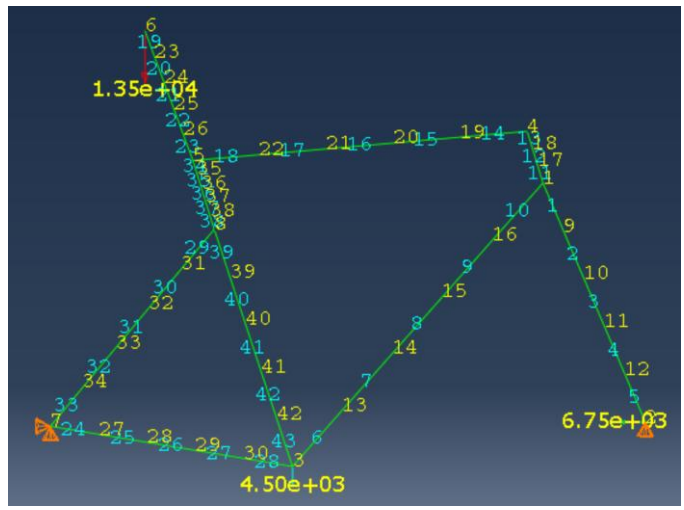

Figure 3 FE geometry with labeled elements and nodes, boundary conditions, and applied loads

The results of 10 iterations of the tube diameters using FSD are represented in tables III and IV. Table III shows the diameter of each tube after every iteration up to 10 and table IV shows the max S11 stress in each tube calculated from the geometry after iterating the diameter which were used to calculate the diameters of the subsequent iteration.

Table III
FSD Iterated Diameter for all Frame Tubes

Iterations	wheel fork	down tube	head tube	top tube
0	60.000	60.000	60.000	60.000
1	69.698	59.110	45.251	47.121
2	70.140	62.964	35.800	37.261
3	70.161	66.392	25.230	27.452
4	70.163	68.117	17.800	19.816
5	70.162	68.223	19.240	15.910
6	70.162	68.670	21.224	12.755
7	70.162	69.083	22.720	10.000

8	70.162	69.289	23.447	10.000
9	70.162	69.335	23.557	10.000
10	70.162	69.339	23.569	10.000
	seat post	chain stays	seat stays	seat tube
0	60.000	60.000	60.000	60.000
1	55.065	29.619	48.275	66.486
2	54.609	14.422	39.757	70.879
3	54.560	10.000	32.671	72.032
4	54.557	10.000	27.447	72.256
5	54.557	10.000	23.550	72.095
6	54.556	10.000	20.862	71.905
7	54.556	10.000	19.068	71.795
8	54.556	10.000	17.884	71.738
9	54.556	10.000	17.111	71.754
10	54.556	10.000	16.604	71.757

Table IV
Calculated Max S11 Stress (FSD Iterations)

Iterations	wheel fork	down tube	head tube	top tube
0	328.436	232.578	132.082	143.992
1	243.185	274.032	144.979	145.027
2	240.15	268.17	110.985	122.874
3	240.013	253.222	107.955	114.763
4	239.994	240.782	287.73	143.4
5	240	243.295	300.855	140.272
6	240	243.024	280.307	121.777
7	240	241.494	257.745	110.203
8	240	240.329	242.543	107.306
9	240	240.033	240.285	106.845
10	240	239.995	239.989	106.772
	seat post	chain stays	seat stays	seat tube
0	200.324	52.708	151.595	297.61
1	235.823	45.7668	158.451	274.302
2	239.547	61.29	156.776	248.213
3	239.966	90.26	163.47	241.56
4	240	95.9	170.295	238.887
5	239.995	98.71	182.127	238.685
6	239.998	100.379	195.036	239.23
7	240.001	101.554	206.677	239.606
8	240	102.213	216.345	240.115
9	240	102.746	223.502	240.018
10	240	103.125	228.707	239.996

After iterating the diameter 10 times using the FSD approach, the final diameters shown in iteration 10 in table III resulted in optimized max S11 stresses all under 240 MPa with all tubes except for seat stays being fully optimized as close to 240 S11 stress as they can be or as small as the tube can be in diameter (10 mm). The deformed final geometry of the bike frame after 10 FSD iterations is shown in figure 4.

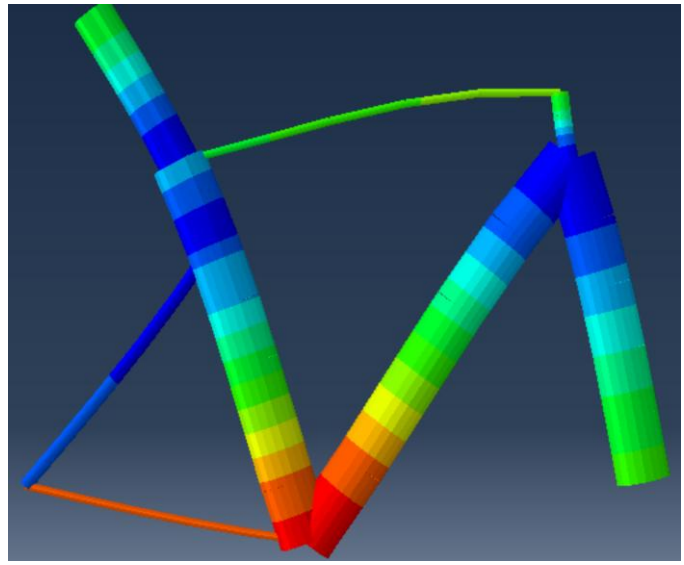

Figure 4 Plot of Deformed Bike Frame Geometry

After doing the 10 iterations of the bike frame tube diameters, the optimized tube diameters were checked for buckling using a buckling analysis. The first buckling analysis with the frame from FSD iterations was analyzed and the first buckling mode eigenvalue was found to be 0.5467 showing that the first buckling mode was the seat stays buckling as shown in figure 5.

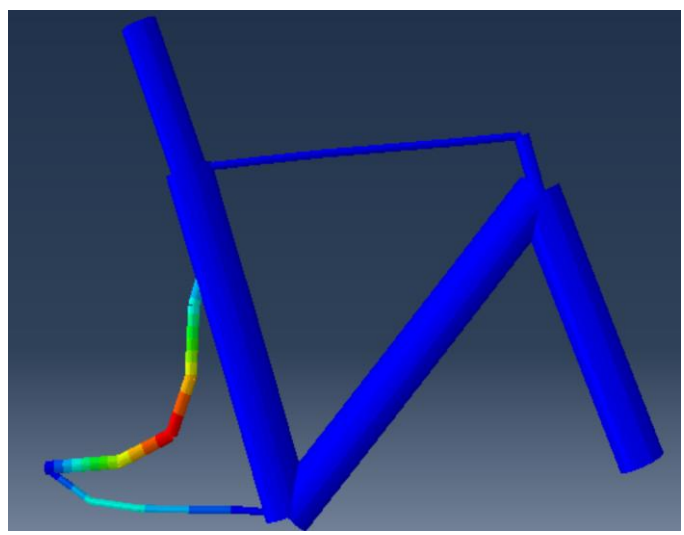

Figure 5 First Buckling Mode Case

Since the eigenvalue was under 1, the seat stay diameter was increased and the buckling analysis was rerun. Table V shows the iterations for the buckling analysis as the diameter of the tube that buckled in the first mode shape in each analysis was heuristically increased until the eigenvalue of the first mode was over 1.

Table V
Heuristic Increase in Diameter from Buckling Analysis

Eigenvalue	Top Tube	Seat Stay
0.5467	0	0
0.66193	0	0.4
0.7271	0	0.5
0.81525	1	0
0.97758	0	0.5
1.0451	0	0.2
Total Increase:	1	1.6

After the top tube and seat stay tube diameters were increased from the buckling analysis data until the eigenvalue was greater than 1 the final values for the diameter of each bike frame tube are shown in Table VI.

Table VI
Final Outer Diameters of Each Frame Tube

Tube	Final Outer Diameters
wheel fork	70.16 mm
down tube	69.34 mm
head tube	23.57 mm
top tube	11 mm
seat post	54.56 mm
chain stays	10 mm
seat stays	18.2 mm
seat tube	71.76 mm

The final optimized bike frame is able to withstand the given load case with the dynamic load factor and safety factor considered as shown in the bending moment diagram in figure 6 and the final mass of the bike frame was 2.04 kg.

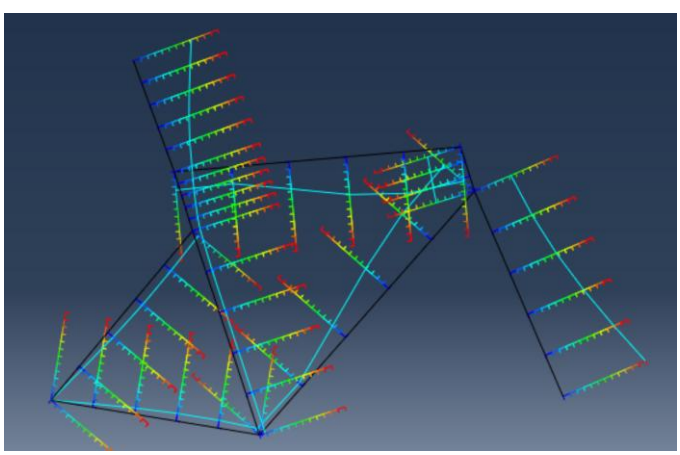

Figure 6 Final Bending Moment Diagram of Bike Frame

IV. Discussion

The objective of this project was to minimize the weight of a bicycle frame by optimizing the diameters of eight aluminum tubular members while ensuring that stresses remain below the material yield strength and that the structure is safe from buckling. A finite element model of the frame was created in Abaqus and subjected to combined vertical and horizontal loads representing rider weight and an impact scenario. Using the fully stressed design (FSD) method, the tube diameters were iteratively adjusted based on the maximum beam stress (S11) until the stresses approached the aluminum yield strength of 240 MPa while maintaining the minimum allowable diameter of 10 mm. After ten iterations, the frame reached a configuration where most members were operating close to the allowable stress limit, indicating an efficient distribution of material throughout the structure.

The FSD results demonstrate that several tubes experienced significantly different stress demands depending on their role within the frame structure. Tubes such as the wheel fork, down tube, seat tube, and seat post retained relatively large diameters because they carried a substantial portion of the applied loads and bending moments. These members approached the allowable stress of 240 MPa after optimization, indicating that the FSD procedure effectively utilized the available material strength. In contrast, tubes such as the chain stays and top tube were reduced to the minimum allowable diameter of 10 mm, suggesting that these members were not heavily stressed under the given load cases. The seat stays remained slightly below the fully stressed condition even after multiple iterations because they were constrained by stability requirements rather than purely by material strength. Overall, the FSD method successfully redistributed material to the most highly loaded regions of the frame, resulting in an efficient structure with a final mass of approximately 2.04 kg.

Although the FSD procedure produced a design that satisfied the yield stress constraint, the subsequent buckling analysis revealed that structural instability governed the design of certain members. The first buckling analysis produced an eigenvalue of 0.5467, which indicates that the frame would buckle at approximately 54.7% of the applied load. The corresponding mode shape showed that the seat stays were the primary members experiencing instability. This result is consistent with structural mechanics principles, since slender compression members are often susceptible to buckling before reaching their yield stress. To address this issue, the diameters of the seat stays and top tube were incrementally increased through a heuristic process until the first eigenvalue exceeded 1.0. After these adjustments, the first eigenvalue increased to approximately 1.045, indicating that the frame can support the applied load without buckling. This result highlights the importance of considering both material strength and structural stability when optimizing lightweight structures.

Despite producing a feasible design, several limitations exist in the analysis. The frame tubes were modeled as straight hollow beams with constant thickness and idealized joints, which neglects the effects of weld geometry, local stress concentrations, and joint stiffness. Additionally, the analysis

assumed linear elastic material behavior and used a static load case multiplied by dynamic and safety factors to approximate real-world loading conditions. In practice, bicycle frames experience complex dynamic loads, fatigue effects, and three-dimensional interactions that are not captured in this simplified model. Therefore, the optimized design obtained in this study should be considered applicable primarily to simplified beam-frame models under similar loading assumptions rather than a complete representation of a real bicycle frame.

V. Conclusion

In this project, a finite element model of a bicycle frame was developed and optimized using a fully stressed design approach combined with buckling analysis. The FSD procedure successfully redistributed material throughout the frame, reducing unnecessary tube diameters while ensuring that stresses remained below the aluminum yield strength of 240 MPa. The resulting design significantly reduced structural weight while maintaining strength requirements. However, the initial optimized design was found to be susceptible to buckling, with the first eigenvalue below 1. By increasing the diameters of the seat stay and top tube members, the buckling resistance of the frame was improved until the first eigenvalue exceeded 1, indicating structural stability under the applied loads. The final optimized frame achieved a mass of approximately 2.04 kg while satisfying both strength and buckling constraints, demonstrating the effectiveness of combining FSD optimization with buckling stability analysis in lightweight structural design.